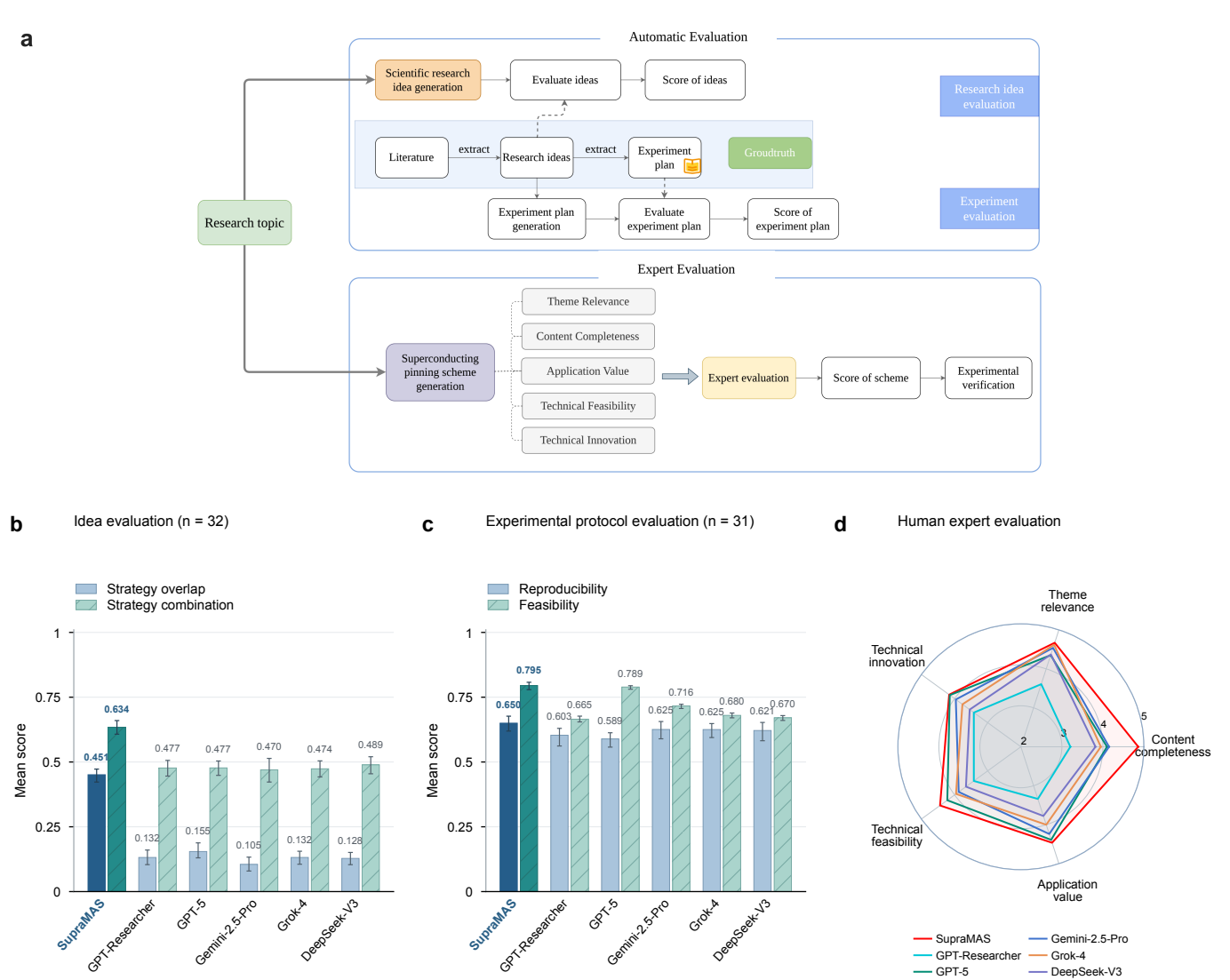

a
Automatic Evaluation
Scientific research idea generation
Evaluate ideas
Score of ideas
Research idea evaluation
Literature
extract
Research ideas
extract
Experiment plan
Groudtruth
Research topic
Experiment plan generation
Evaluate experiment plan
Score of experiment plan
Experiment evaluation
Expert Evaluation
Theme Relevance
Content Completeness
Superconducting pinning scheme generation
Application Value
Expert evaluation
Score of scheme
Experimental verification
Technical Feasibility
Technical Innovation
b
Idea evaluation (n = 32)
Strategy overlap
Strategy combination
Mean score
1
0.75
0.5
0.25
0
0.451
0.634
0.132
0.477
0.155
0.477
0.105
0.470
0.132
0.474
0.128
0.489
SupraMAS
GPT-Researcher
GPT-5
Gemini-2.5-Pro
Grok-4
DeepSeek-V3
c
Experimental protocol evaluation (n = 31)
Reproducibility
Feasibility
Mean score
1
0.75
0.5
0.25
0
0.650
0.795
0.603
0.665
0.589
0.789
0.625
0.716
0.625
0.680
0.621
0.670
SupraMAS
GPT-Researcher
GPT-5
Gemini-2.5-Pro
Grok-4
DeepSeek-V3
d
Human expert evaluation
Theme relevance
Content completeness
Application value
Technical feasibility
Technical innovation
2
3
4
5
SupraMAS
GPT-Researcher
GPT-5
Gemini-2.5-Pro
Grok-4
DeepSeek-V3